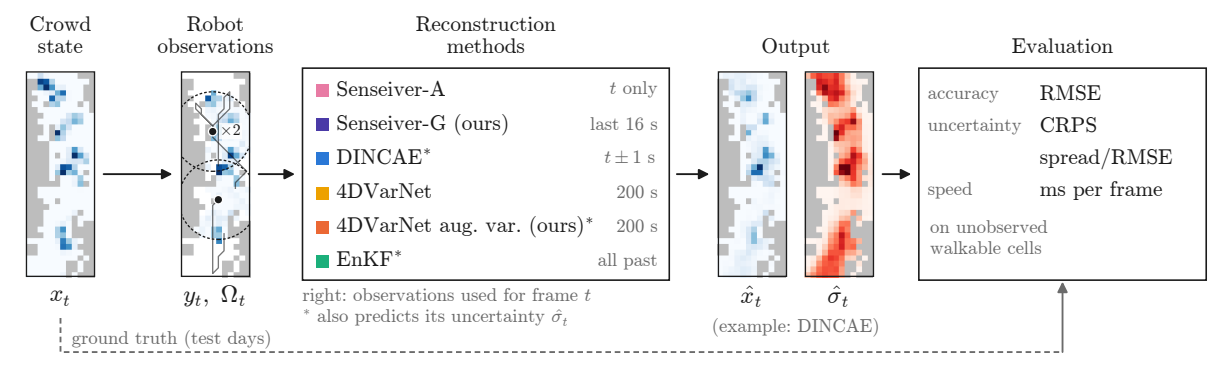
Crowd state
Robot observations
Reconstruction methods
Output
Evaluation
Senseiver-A
t only
Senseiver-G (ours)
last 16 s
DINCAE*
t ± 1 s
4DVarNet
200 s
4DVarNet aug. var. (ours)*
200 s
EnKF*
all past
accuracy
RMSE
uncertainty
CRPS
spread/RMSE
speed
ms per frame
on unobserved walkable cells
×2
x_t
$y_t,\ \Omega_t$
right: observations used for frame t
* also predicts its uncertainty $\hat{\sigma}_t$
$\hat{x}_t$
$\hat{\sigma}_t$
(example: DINCAE)
ground truth (test days)